

MAT 426, Class 8 Homework: Orthogonal Basis & Vector Projections
(Due Thursday, March 5, 2026)

1. Determine which sets of vectors are orthogonal.

(a) $\left\{ \begin{bmatrix} -1 \\ 4 \\ -3 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -4 \\ -7 \end{bmatrix} \right\}$

(c) $\left\{ \begin{bmatrix} 5 \\ -4 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} -4 \\ 1 \\ -3 \\ 8 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 5 \\ -1 \end{bmatrix} \right\}$

(b) $\left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix} \right\}$

(d) $\left\{ \begin{bmatrix} 3 \\ -2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ 8 \\ 7 \\ 0 \end{bmatrix} \right\}$

2. Show that $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} \right\}$ is an orthogonal basis for $\mathbb{R}^3$ and then express $\begin{bmatrix} 8 \\ -4 \\ -3 \end{bmatrix}$ as a linear combination of the basis elements.

3. Show that $\left\{ \begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} \right\}$ is an orthogonal basis for $\mathbb{R}^3$ and then express $\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}$ as a linear combination of the basis elements.

4. Let $\mathbf{y} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$. Write $\mathbf{y}$ as the sum of two orthogonal vectors, one in $\text{Span}\{\mathbf{u}\}$ and one orthogonal to $\mathbf{u}$.

5. Compute the orthogonal projection of $\begin{bmatrix} 1 \\ 7 \end{bmatrix}$ onto the line through $\begin{bmatrix} -4 \\ 2 \end{bmatrix}$ and the origin.

6. Determine which of the following sets of vectors are orthonormal. If a set is only orthogonal, normalize the vectors to produce an orthonormal set.

(a) $\left\{ \begin{bmatrix} -2/3 \\ 1/3 \\ 2/3 \end{bmatrix}, \begin{bmatrix} 1/3 \\ 2/3 \\ 0 \end{bmatrix} \right\}$

(b) $\left\{ \begin{bmatrix} 1/\sqrt{10} \\ 3/\sqrt{20} \\ 3/\sqrt{20} \end{bmatrix}, \begin{bmatrix} 3/\sqrt{10} \\ -1/\sqrt{20} \\ -1/\sqrt{20} \end{bmatrix}, \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \right\}$

(c) $\left\{ \begin{bmatrix} 1/\sqrt{18} \\ 4/\sqrt{18} \\ 1/\sqrt{18} \end{bmatrix}, \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}, \begin{bmatrix} -2/3 \\ 1/3 \\ -2/3 \end{bmatrix} \right\}$

7. Suppose $\mathbf{u} \neq \mathbf{0}$ in $\mathbb{R}^n$ and let $L = \text{span}\{\mathbf{u}\}$. For $\mathbf{y}$ in $\mathbb{R}^n$, the reflection of $\mathbf{y}$ in L is the point $\text{refl}_L \mathbf{y}$ defined by $\text{refl}_L \mathbf{y} = 2\text{proj}_L \mathbf{y} - \mathbf{y}$.

- (a) Draw a sketch in $\mathbb{R}^2$ to show that $\text{refl}_L \mathbf{y}$ is the sum of the vectors $\hat{\mathbf{y}} = \text{proj}_L \mathbf{y}$ and $\hat{\mathbf{y}} - \mathbf{y}$.
 (b) Show that the mapping $T(\mathbf{y}) = \text{refl}_L \mathbf{y}$ is a linear transformation.